SCEPI: Single-Mutant-Conditioned Epistasis Prediction for Compensatory Protein Variant Discovery

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Abstract. We introduce Single-mutant-Conditioned Epistasis Prediction (SCEPI) for discovering compensatory double mutants from individually damaging substitutions. A shared cross-protein interaction function receives measured single-mutant effects alongside protein representations. The same measurements calibrate individual effects outside the interaction network. Training uses source combination measurements and withholds target double-mutant labels. Across 155 held-out proteins, SCEPI achieves mean residual root mean squared error of 0.595 kcal/mol, compared with 0.805 for ThermoMPNN-D, 0.847 for Kermut, and 0.853 for calibrated StabilityArc. Selecting ten candidates per protein yields a compensation hit rate of 11.22% across 120 proteins, compared with 7.07%, 5.40%, and 8.75%, respectively. This is 4.27 times the 2.63% expectation of uniform random selection from the same candidates. With external calibration held fixed, matched input-removal controls identify prediction and selection gains from supplying measured effects to the interaction network. These gains persist with the same measured single-effect sum, stricter homology grouping, and measurement-boundary analysis. Existing single-mutant measurements thus guide compensatory discovery in unseen proteins through a transferable interaction model.

1 Introduction

Compensatory substitutions recover stability or function lost through other sequence changes [1, 2, 5]. Discovering them requires selecting useful combinations of individually damaging substitutions. Single-mutant measurements provide target-specific information for spending a fixed experimental budget on these combinations.

Complete wild-type, single-mutant, and double-mutant measurements define observed nonadditivity [13]. This contrast includes global phenotype nonlinearity and pair-dependent effects [10, 11]. Measurements from source proteins can supervise a shared interaction function for predicting unmeasured target combinations. Transferable combination predictors already learn from structure or sequence representations [3, 9, 14], and target-specific regression uses measured variants [6]. We ask whether target single-mutant measurements add predictive value when supplied directly to the shared interaction function.

SCEPI uses measured single effects in two places. They calibrate individual effects and condition a shared interaction function together with their observation indicators. Within this function, the measured inputs accompany reference and single-mutant representations. Subtraction around a double-mutant cycle cancels endpoint-only offsets and enforces zero interaction at either wild-type endpoint. Source combination measurements train the shared parameters; target single-mutant observations condition predictions at inference.

Under both homology groupings, measured inputs improve residual prediction and compensation selection. The input-removal controls retain external calibration, model capacity, training samples, and selection procedures. In the full-target comparison, SCEPI's top-10 hit rate is 11.22%, compared with 7.07% for ThermoMPNN-D, 5.40% for Kermut, and 8.75% for calibrated StabilityArc. Every method receives zero target double-mutant training labels. The selection gains persist after fixing the measured single-effect sum and excluding measurements near assay limits. A separate same-capacity direct-head comparison tests the reference constraints.

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2 Related work

Combination predictors from protein representations. Transferable predictors use structural or sequence representations to model combinations. In ThermoMPNN-D, frozen ProteinMPNN features feed a Siamese head [3]. Mutate Everything composes reusable mutation representations [9], and ESM-MSR encodes wild-type and mutant sequences [12]. The StabilityArc preprint learns a shared wild-type decoder with a symmetric, contact-conditioned residual [14].

Using target measurements. Single-mutant labels enter combination predictors at different stages (Table 1). Mutate Everything uses measured singles to construct training residuals, then supplies predicted effects at inference [9]. For target adaptation, StabilityArc provides a cross-protein prior to Kermut's Gaussian-process posterior [6, 14]. In SCEPI, measured effects enter both external calibration and the shared pair network. Holding the calibration fixed during input removal isolates the contribution of their use within that network.

Phenotype maps and interaction terms. MAVE-NN, MoCHI, and reference-free analysis fit genetic effects through nonlinear phenotype maps [4, 11, 15]. Global epistasis distinguishes this nonlinearity from pair-dependent contributions [10]. Both enter SCEPI's predicted phenotype contrast. We evaluate the map and fitting objectives separately from measured-input conditioning.

3 Methods

Source combination measurements train SCEPI's shared interaction function (Figure 1). At inference, target single effects condition this function and calibrate the external single-effect sum. Target double-mutant outcomes are reserved for evaluation.

3.1 Prediction problem

Let s_p be a reference protein and M a set of substitutions at distinct sites. The predictor receives s_p, M, and the target measurements allowed by the information condition. It outputs the relative phenotype $\widehat{\mu}_p(M)$ and a candidate score. For two substitutions $A = (i, a)$ and $B = (j, b)$, the measured interaction is

$$\epsilon_p(A, B) = y_p(AB) - y_p(A) - y_p(B) + y_p(\varnothing). \tag{1}$$

The four observations form a double-mutant cycle [7]. They share a reference background, assay condition, and phenotype scale. Stability is expressed as unfolding free energy, with positive changes indicating increased stability. Target single-mutant labels are available in the main compensation task. Target test double-mutant labels evaluate predictions and selected candidates.

Table 1. Single-mutant information use. The final column records direct measured-value inputs to the pair network. ThermoMPNN-D target-head fitting refers to our single-only target protocol. Mutate Everything is included for the methodological comparison.

Method	Source use of single labels	Target-single use	Measured pair inputs
ThermoMPNN-D [3]	Data augmentation	Head adaptation	No
Mutate Everything [9]	Residual targets	Predicted effects	No
StabilityArc + Kermut [6, 14]	Decoder supervision	Target GP posterior	No
SCEPI	Joint loss	Calibration and conditioning	Effects and masks

3.2 Mutation representations and calibrated single effects

A frozen ESM-2 encoder h supplies the sequence representations [8]. For substitution $A = (i, a)$, the endpoint representation contains $h_i(s_p)$, the local change $h_i(x_p(A)) - h_i(s_p)$, amino-acid identities, and the pretrained substitution score. The single-effect head is shared across proteins and uses a reference subtraction,

$$a_i(a) = f_\phi(u_i(a)) - f_\phi(u_i(w_i)), \tag{2}$$

where w_i is the reference residue, giving $a_i(w_i) = 0$.

Available single-mutant effects enter through

$$\widetilde{a}_i(a) = a_i(a) + \omega m_i(a)\big[\mu_i^{\mathrm{obs}}(a) - a_i(a)\big], \tag{3}$$

where $m_i(a)$ records label availability and ω is selected on source-validation proteins. An observed zero remains an available label. The calibrated effects enter both the single-effect sum and the interaction network. Pair representations also include mutation-dependent cross-site context and sequence separation, defined with the structural variant in Appendix A.

3.3 Conditioning the shared interaction function

For a candidate pair, collect the calibrated single effects and their availability indicators in

$$c_{ij}(a, b) = \big(\widetilde{a}_i(a), \widetilde{a}_j(b), m_i(a), m_j(b)\big). \tag{4}$$

Together with the endpoint representations and site relation, c_{ij} makes the shared network depend on measured effects from the target protein. Exchanging the endpoints also exchanges the entries of c_{ij}.

Let $\Psi_\theta(i, a; j, b \mid c_{ij})$ denote the ordered network. We average both orders:

$$F_\theta(i, a; j, b) = \tfrac{1}{2}\big[\Psi_\theta(i, a; j, b \mid c_{ij}) + \Psi_\theta(j, b; i, a \mid c_{ji})\big]. \tag{5}$$

The conditioning inputs enter every evaluation of F. The pair output uses the same subtraction over the four states,

$$r_{ij}(a, b) = F(i, a; j, b) - F(i, a; j, w_j) - F(i, w_i; j, b) + F(i, w_i; j, w_j). \tag{6}$$

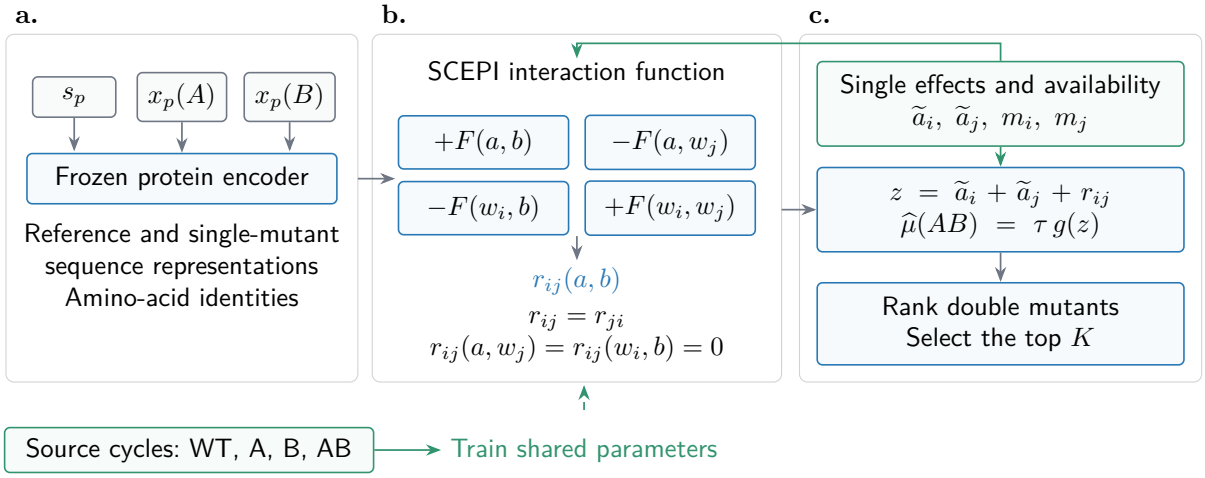


Fig. 1. SCEPI conditions shared interaction predictions on single-mutant measurements. (a) A frozen encoder supplies reference and mutation features from the reference sequence and its single mutants. (b) Calibrated single effects and their availability indicators condition every evaluation of the shared symmetric pair function. Subtraction around the double-mutant cycle enforces the reference constraints. (c) The same single effects enter the external sum before phenotype prediction. Double-mutant predictions then rank candidates at a fixed budget. Complete source cycles and source observations supervise SCEPI. Target double-mutant outcomes are reserved for evaluation.

Each call evaluates the specified amino-acid state. At a wild-type endpoint, the single effect and local representation change are zero. The context change caused by that endpoint at the other site is also removed.

The subtraction enforces symmetry, $r_{ij}(a,b) = r_{ji}(b,a)$, and zero reference effects, $r_{ij}(a,w_j) = r_{ij}(w_i,b) = 0$. It also cancels every endpoint-only contribution, so replacing F by $F + v_i(a) + v_j(b) + c$ leaves r unchanged. We test the predictive contribution of these constraints against a matched direct head.

3.4 Composition and phenotype prediction

For a mutation set M, the latent effect sums single and pair contributions:

$$z_p(M) = \sum_{(i,a)\in M} \widetilde{a}_i(a) + \sum_{\substack{(i,a),(j,b)\in M \\ i<j}} r_{ij}(a,b). \tag{7}$$

We predict $\widehat{\mu}_p(M) = \tau g(z_p(M))$, where $\tau > 0$, $g(0) = 0$, and $g'(0) = 1$. The monotone map models global phenotype nonlinearity. Its parameterization is given in the appendix.

The model's interaction on the observed phenotype scale is always calculated from its four predictions:

$$\widehat{\epsilon}_p(A,B) = \widehat{\mu}_p(AB) - \widehat{\mu}_p(A) - \widehat{\mu}_p(B). \tag{8}$$

This contrast includes global phenotype nonlinearity and the learned pair contribution. It equals r_{ij} under the identity map and unit scale. Appendix A gives the decomposition.

3.5 Learning from complete combination measurements

After initialization on source single-mutant measurements, the single head is fitted jointly with the pair function and phenotype map using source combinations. The objective combines a Huber loss on the complete predicted residual with the likelihood of point-valued and interval-valued source observations. Sampling balances proteins and site pairs.

To represent inference conditions, source training varies the permitted single-mutant information. Model configurations and the label-mixing weight are selected on source-validation proteins, excluding target homology groups from fitting and selection. Appendix C gives the criterion, observation model, and implementation. Residual-only and phenotype-only variants test the choice of objective.

3.6 Compensatory candidate selection

For a budget K, SCEPI selects the K double mutants with the highest native phenotype predictions $\widehat{\mu}_p(AB)$. Recovery is measured relative to the better constituent single mutant,

$$R_p(A,B) = y_p(AB) - \max\{y_p(A), y_p(B)\}. \tag{9}$$

A compensatory hit satisfies $R_p(A,B) \geq \delta$. The candidate task focuses on pairs whose two measured single effects are each at most $-\delta$.

To isolate interaction information, we fix the measured single-effect sum across methods. Each method m ranks the same candidates by

$$s_m(A,B) = \mu_p^{\mathrm{obs}}(A) + \mu_p^{\mathrm{obs}}(B) + \widehat{\epsilon}_{m,p}(A,B). \tag{10}$$

Each method retains its own four-prediction contrast, so scores differ through predicted nonadditivity. Native ranking evaluates the complete predictor; Equation (10) isolates the selection value of its interaction predictions.

Uniform random selection samples K candidates without replacement from the same protein-level pool. Its expected hit rate is the pool's hit fraction, and its expected mean recovery is the pool mean. Enrichment is the ratio of the model and random equal-protein mean hit rates. Appendix K gives the exact calculation and uncertainty analysis.

4 Experimental setting

The cDNA-display proteolysis measurements of Tsuboyama et al. provide stability data [13]. Among 182,663 accepted double mutants across 190 reference backgrounds, 127,892 form complete, point-valued cycles across 160 backgrounds and 488 site pairs. Retrospective evaluation requires all four measurements to lie inside the assay range of $[-1, 5]$ kcal/mol. Training additionally retains interval-valued observations through the observation likelihood.

In the primary S-all evaluation, all permitted target single-mutant labels are available and every target double-mutant label is withheld from fitting. We score all accepted double mutants in held-out homology groups. Groups use 30% sequence identity and at least 80% bidirectional coverage, keeping related reference backgrounds together. Annotated family relations further merge groups. Source training and validation exclude the outer test partition. Coverage is recorded separately for homology groups and annotated families.

We compare independently trained additive and global nonlinear models with ThermoMPNN-D, Kermut, StabilityArc + Kermut, mean ridge, and mean MLP. The global control keeps the monotone map and omits the pair term. Mean ridge and mean MLP fit ridge regression or a multilayer perceptron (MLP) to averaged mutation representations. For source-trained baselines, retraining uses the same source partitions and source-validation checkpoint selection. StabilityArc scores provide Kermut's prior mean during target calibration [6, 14]. All methods score the same candidates with zero target double-mutant labels. Their architectures, supervision, adaptation, and structural inputs are specified in Appendices C and D.

Prediction error is measured by root mean squared error (RMSE). The phenotype cohort contains 159 proteins from 51 homology groups, each with at least 30 point-valued target measurements. Residual evaluation requires at least 30 complete, in-range cycles and retains 155 proteins from the same 51 groups. For compensation selection, we require at least 100 eligible candidates and reference stability at most 4.5 kcal/mol. This gives 120 proteins from 44 groups. Candidate budgets are $K \in \{1, 3, 10, 30\}$, with recovery threshold $\delta = 0.5$ kcal/mol.

Metrics are computed within protein, averaged over matched repetitions, and then averaged equally across proteins. Source-validation measurements determine hyperparameters and label-mixing weights.

Measurement-boundary analysis uses each cycle's smallest distance from either assay limit. Margins of 0.25 and 0.5 kcal/mol retain 120 and 115 compensation proteins, respectively; the latter covers 43 groups. The designated site-pair subset and target combination adaptation are evaluated separately. Statistical and repetition details appear in the appendix.

We vary target-label availability in additional information conditions. Z permits no target measurements; S32 and S128 permit the reference and the specified number of single-mutant labels. The C conditions also permit 128, 512, 2,048, or all available training double-mutant labels, excluding held-out test site pairs from target combination fitting. Separate adaptation experiments cover six functional assays and nine FLIP2 split protocols [17]. Appendix I defines their measurement units, model variants, and evaluation populations.

5 Results

5.1 Measured single effects improve interaction prediction and selection

Measured single effects within the pair function reduce residual RMSE by 22.4% under the primary grouping and 25.7% under the stricter grouping (Table 2; Figure 4c,d). Native top-10 hit rates rise by 3.03 and 3.00 percentage points. Both mean gains are positive in every matched training repetition under each grouping. To isolate this input contribution, the removal control supplies model-predicted single effects and zero availability indicators to F, retaining external calibration and its source-validation selection procedure. Each model is refitted independently with matched capacity, initialization, training samples, objectives, and stopping rules. Controls retain the same 155 residual-evaluation proteins and 120 compensation proteins.

Fixing the measured single-effect sum preserves hit-rate gains of 3.08 and 3.02 percentage points over the input-removal control. At the 0.5 kcal/mol measurement margin, these gains are 3.11 and 3.13 points. The pair network therefore uses measured effects to improve selection even when the external single-effect sum is identical.

Table 2. Matched pair-function comparisons under 30% and 25% homology grouping. Residual prediction covers 155 proteins, and native top-10 selection covers 120. The predicted-singles control removes measured values and availability indicators from F while retaining external calibration. The direct head keeps those inputs and replaces cycle subtraction with the same network's direct output.

	Residual RMSE (kcal/mol)		Native hit rate (%)	
Pair function	30%	25%	30%	25%
SCEPI	**0.595**	**0.604**	**11.22**	11.15
Predicted singles	0.767	0.814	8.18	8.15
Direct residual head	0.635	0.630	10.70	**11.38**

Table 3. Full-target prediction and native top-10 compensation selection in held-out homology groups. Phenotype, residual, and selection cohorts contain 159, 155, and 120 proteins. RMSE uses kcal/mol. Values are equal-protein means over matched repetitions. Enrichment is relative to uniform random selection from the same candidates. All models receive zero target double-mutant training labels.

Predictor	Phenotype RMSE	Residual RMSE	Hit rate (%)	Recovery (kcal/mol)	Enrichment (×)
SCEPI	**0.657**	**0.595**	11.22	−0.257	4.27
ThermoMPNN-D	0.777	0.805	7.07	−0.440	2.69
Kermut	0.798	0.847	5.40	−0.458	2.05
StabilityArc + Kermut	0.815	0.853	8.75	−0.344	3.33
Global nonlinear	0.911	0.635	4.45	−0.437	1.69
Additive	0.893	1.091	6.38	−0.387	2.43
Mean MLP	1.405	0.966	10.07	−0.413	3.83
Mean ridge	1.328	1.002	**11.42**	−0.354	**4.34**
Uniform random	—	—	2.63	−0.900	1.00

5.2 The conditional predictor improves compensatory candidate discovery

At a budget of $K = 10$ candidates, native SCEPI predictions yield an 11.22% mean compensation hit rate (Table 3). ThermoMPNN-D reaches 7.07%, Kermut 5.40%, and the global nonlinear model 4.45%. The corresponding hit-rate gains are 4.15, 5.82, and 6.77 percentage points, with mean recovery gains of 0.183, 0.201, and 0.180 kcal/mol. Higher recovery occurs in 86 of 120 proteins against ThermoMPNN-D and 93 of 120 against Kermut. StabilityArc + Kermut reaches an 8.75% top-10 hit rate. Against this method, SCEPI gains 2.47 percentage points in hit rate and 0.087 kcal/mol in mean recovery, with higher recovery in 73 of 120 proteins. Both predictors learn pair effects across source proteins and use target single-mutant measurements. Figure 2 shows the budget curves and all protein-level hit-rate differences.

Mean selected recovery favors SCEPI over all seven baselines at every evaluated budget. At $K = 10$, mean ridge has a slightly higher hit rate of 11.42%, compared with SCEPI's 11.22% and mean MLP's 10.07%. SCEPI's recovery advantage over mean ridge is 0.098 kcal/mol. Relative to the additive model's 6.38% hit rate, SCEPI gains 4.83 percentage points and 0.130 kcal/mol mean recovery.

Uniform random selection from the same 79,564 candidates across 120 proteins has an expected equal-protein hit rate of 2.63% and mean recovery of −0.900 kcal/mol. At $K = 10$, SCEPI enriches hits 4.27-fold, gaining 8.59 percentage points in hit rate and 0.643 kcal/mol in mean recovery. Enrichment decreases from 5.52 to 4.90, 4.27, and 3.26 as the budget increases through 1, 3, 10, and 30 candidates (Figure 2a,b).

5.3 The learned function predicts interactions in unseen combinations

Residual RMSE falls to 0.595 kcal/mol with SCEPI, compared with 0.805 for ThermoMPNN-D and 0.847 for Kermut. These are reductions of 26.1% and 29.8%, with lower error in 136 of 155 proteins against either method (Figure 3a). Against StabilityArc + Kermut's 0.853 kcal/mol, the reduction is 30.3%, and error is lower in 130 of 155 proteins.

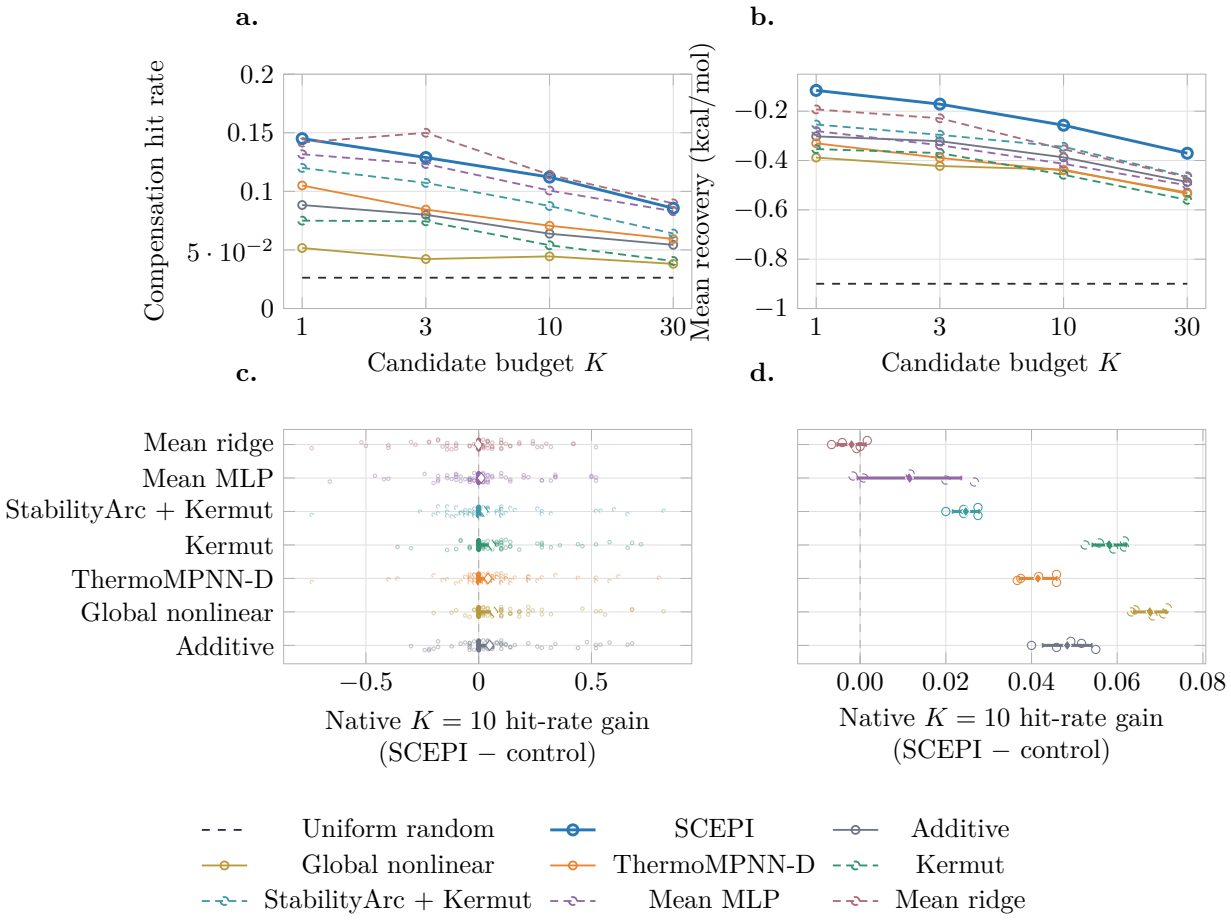


Fig. 2. SCEPI compensatory discovery in full-target S-all evaluation. (a) Native hit rate and (b) mean recovery across candidate budgets. Dashed black lines show exact uniform-random expectations from the same candidates. (c) Protein-level top-10 hit-rate gains after averaging repetitions within each of 120 proteins. Open points denote proteins; segments, ticks, and diamonds denote interquartile ranges, medians, and means. (d) Repetition-level gains in the same control order. Open points are equal-protein means; diamonds and whiskers show their mean and standard deviation. All predictors use identical candidates and native phenotype scores. StabilityArc + Kermut uses source-held-out scores with target calibration.

Global phenotype nonlinearity reduces the additive model's residual error from 1.091 to 0.635 kcal/mol. Adding the learned pair function lowers this error by another 6.4%, with improvement in 92 of 155 proteins. The two comparisons distinguish the contributions of the phenotype map and the pair function.

For phenotype prediction, SCEPI reaches RMSE 0.657 kcal/mol and mean Spearman correlation 0.742. The corresponding values are 0.777 and 0.673 for ThermoMPNN-D, 0.798 and 0.696 for Kermut, and 0.815 and 0.746 for StabilityArc + Kermut. Protein-level phenotype error is lower in 117 of 159 comparisons with ThermoMPNN-D and 114 of 159 with Kermut. Both target-fitted mean-representation baselines also have higher mean phenotype and residual errors. Protein-level differences and repetition-level residual gains appear in Figure 3b–d.

Providing 128 target double-mutant training labels allows adaptation to the target protein. On unseen site pairs, the source-initialized SCEPI variant reaches RMSE 0.567 kcal/mol across 70 proteins. Kermut reaches 0.758, ThermoMPNN-D 0.775, second-order MoCHI 0.859, and the target-only SCEPI variant 0.799. Results for every comparator across the four budgets appear in Table 13.

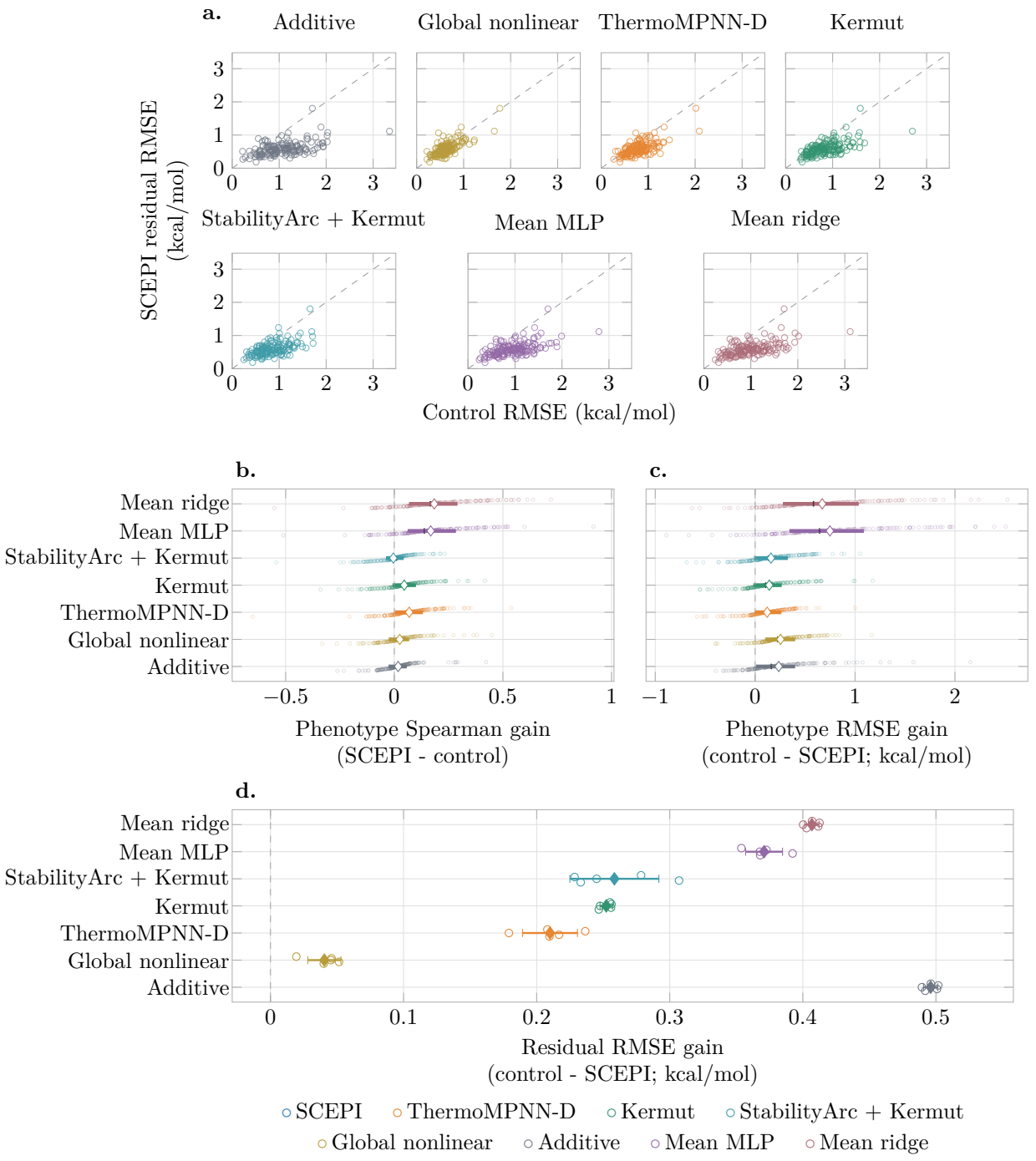


Fig. 3. SCEPI prediction across held-out homology groups in the full-target S-all setting. (a) Paired residual RMSE for all 155 eligible proteins. The dashed diagonal denotes equal error. (b) Native phenotype Spearman gain across 159 proteins. (c) Phenotype RMSE gain in the same cohort. In (b) and (c), points are proteins, segments are interquartile ranges, dark ticks are medians, and diamonds are means. (d) Repetition-level residual RMSE gains. Open points are equal-protein macro averages; diamonds and whiskers show the mean and standard deviation. Positive gains favor SCEPI. All eligible proteins are retained.

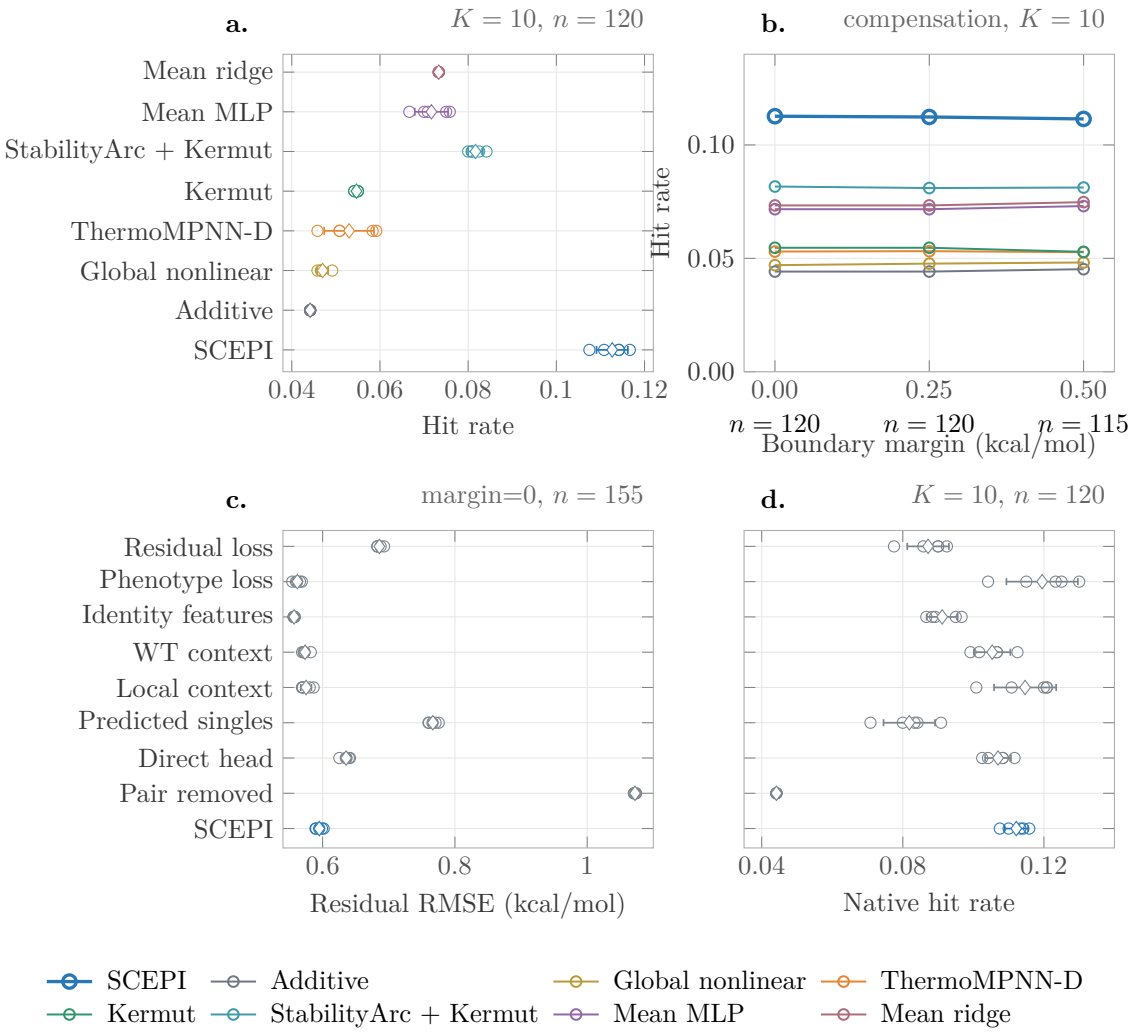


Fig. 4. Measured-single conditioning and component comparisons under 30% identity grouping. (a) Top-10 hit rates with the same measured single-effect sum. (b) Sensitivity to distance from assay limits; cohort sizes appear below the axis. (c) Residual RMSE and (d) native hit rate share the labeled control order. The direct head has matched capacity. The input-removal control retains external calibration and removes measured effects and availability from F. Open points are equal-protein repetition means; diamonds and whiskers show their mean and standard deviation. Residual and selection cohorts contain 155 and 120 held-out proteins.

5.4 Interaction predictions add value beyond measured single effects

When every method uses the same measured single-effect sum, SCEPI retains an 11.27% compensation hit rate (Figure 4a; Appendix Table 7). Additivity gives 4.42% and global nonlinearity 4.70%; ThermoMPNN-D reaches 5.30%, Kermut 5.47%, and StabilityArc + Kermut 8.17%. Mean MLP and mean ridge reach 7.17% and 7.33%. Recovery gains over the first four controls are 0.176, 0.161, 0.224, and 0.200 kcal/mol, respectively, with a 0.095 kcal/mol gain over StabilityArc + Kermut. Only the predicted four-term nonadditivity differs between paired scores.

Restricting evaluation to measurements at least 0.5 kcal/mol from both assay limits preserves this advantage (Figure 4a,b). Hit rates are 11.15% for SCEPI, 4.52% for additivity, 4.82% for global nonlinearity, 5.27% for ThermoMPNN-D, and 5.29% for Kermut. StabilityArc + Kermut reaches 8.12%, mean MLP 7.30%, and mean ridge 7.48%.

5.5 Reference constraints improve residual prediction

Cycle subtraction enforces the reference constraints and improves residual accuracy against the same-capacity direct head (Tables 8 and 9). The direct head reaches residual RMSE 0.635 kcal/mol under the primary grouping and 0.630 kcal/mol under the stricter grouping, compared with SCEPI's 0.595 and 0.604 kcal/mol. Native hit-rate gains are 0.52 and -0.23 percentage points, respectively. The residual benefit holds across both groupings; the hit-rate effect varies.

Setting the pair term to zero at inference lowers native hit rate from 11.22% to 4.42% and raises residual RMSE from 0.595 to 1.072 kcal/mol (Figure 4c,d). The trained single head and phenotype map remain fixed during this intervention. Complete representation, mapping, and training-objective comparisons are reported in the appendix.

6 Discussion

Single-mutant measurements can guide compensatory discovery before any target combination labels are collected. SCEPI uses them to condition an interaction function learned across source proteins. The retrospective comparisons quantify the resulting selection benefit at a fixed budget and across held-out homology groups. This use of target information extends beyond fitting individual effects or a Gaussian-process posterior. StabilityArc supplies a source-trained prior for target adaptation [14]; SCEPI also supplies observed single effects to the shared interaction function. Holding external calibration fixed during input removal identifies the value of that input location. Selection gains also persist with the same measured single-effect sum.

The reference constraints fix the model's behavior at wild-type endpoints and improve mean residual accuracy against a direct head. Their hit-rate effects vary across groupings. Measured-input conditioning improves both residual prediction and compensatory selection under the two homology groupings.

7 Conclusion

By conditioning a transferable interaction function on measured single effects, SCEPI improves residual prediction and compensatory discovery in matched controls and held-out comparisons. Code, configurations, and reproduction materials are available at `https://github.com/SZU-ADDG/SCEPI`.

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